EE5630 Digital Signal Processing

HW1: Sampling and DTFT

Due Friday Oct. 1, 2021

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The first two problem sets involve manual calculation and derivation only.

1. Let $x[n] = u[n] \cdot A\,\mathrm{e}^{-\beta n}\cos(\omega_0 n)$, where $u[n]$ is the discrete-time unit step function. Using the relation $\cos(\theta) = \frac{1}{2}\left(e^{j\theta} + e^{-j\theta}\right)$, derive the DTFT for $x[n]$.

2. Proof of Parseval's theorem. Assume that $x[n]$ is real for all n.
   (a) Let $y[n] = x[n] * x[-n]$ where * denotes convolution. Show that
   $y[0] = \sum_n |x[n]|^2$.
   (b) Show that $Y\left(e^{j\omega}\right) = \left|X\left(e^{j\omega}\right)\right|^2$.
   (c) Calculate the inverse Fourier transform of $Y\left(e^{j\omega}\right)$ to complete the proof.

The next two problems are designed to get you started with MATLAB notations and basic operations such as loading up and listening to audio files.

3. **Fractional delay – time domain and frequency domain representations.**

(a) Manually calculate an explicit expression for the inverse discrete-time Fourier transform $h[n] = \mathcal{F}^{-1}\{H(\omega)\}$, where $H(\omega) = e^{-j\omega/3}$, for $\omega \in [-\pi, \pi]$.

(b) The result should correspond to a filter that represent 1/3 sample of delay. Use Matlab to plot $h[n]$ for $n = -20$ to 20. Hint: you may set `n=-20:20`. Since MATLAB is really convenient for vector operation, instead of using a for loop, please try writing down a *vector expression*[1] `h = ...` directly as a function of n, and then type `plot(n,h)` or `stem(n,h).` Note that the arrays n and h should have the same size. When in doubt, type `whos` to see the list of variables and their sizes in the current work space.

(c) You may plot the DTFT of $h[n]$ using `H = freqz(h)`. Please check how closely the phase spectrum resembles a straight line, and what approximately

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[1] Here is an example of *vector expression* --- `n = 1:10;`
`x = sin(2*pi*n/10).* exp(-n/20)`; note how conveniently arrays can be passed to numerical functions as arguments. Also note the point-wise operator `.*`

is the slope. Is the slope $-\frac{1}{3}$? Explain why or why not.

4. **Sampling and resampling**. First, load a music file as follows: `[x,fs] = audioread('ABBA-flac2wav.wav');` and listen to it using `sound(x,fs).`

(a) Answer the following questions, and describe what you hear.

i. What does it mean to do `sound(x(2:2:end,:), fs)?`

ii. How about `sound(x(2:2:end,:), fs/2)?`

iii. How about `sound(resample(x,fs/2,fs), fs/2)?`

(b) Can you hear the difference between ii and iii? The difference might be subtle (I frankly cannot really tell). Check that they are indeed mathematically different by plotting y = `x(2:2:end,:) - resample(x,fs/2,fs)`.

(c) Continuing from (b), keep trying a different down-sampling rate (e.g., 3, 4, 5,…) until you can hear the difference.